

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	Wall upgrade: • Exterior wall insulation: Mineral Wool Heavy Density Blanket, target R-30.0 Roof upgrade: • Roof insulation: Graphite Polystyrene (GPS) Foam Board, target R-28.0

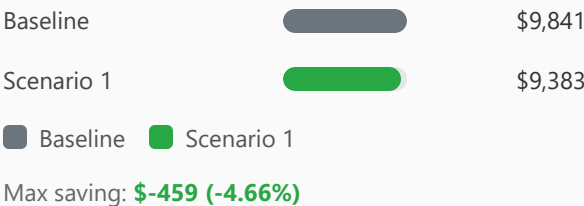
		Window upgrade: • 2-pane replacement (skipped: SimpleGlazing windows) • Weatherstrip application (skipped: no OperableWindow) • Glazing film application (skipped: SimpleGlazing windows) • Caulking application: polyurethane • Window frame replacement: wood-aluminium window frame Door upgrade: • Door replacement: honeycomb core steel door (1 of 3 door(s) skipped: door type incompatibility) • Bottom seal: brush weatherstrip • Top/side seal: jamb weatherstrip
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Annual Energy Analysis

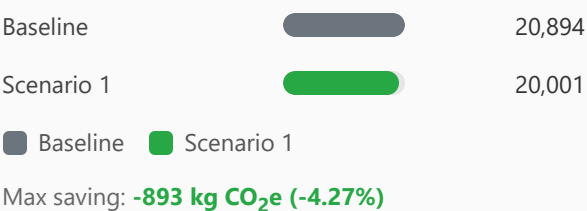
Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	236.54	223.23	13.31	5.63%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison



Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison



Min Retrofit Construction Cost

\$106,899

Min Retrofit Embodied Carbon

21,316 kg CO₂e

Scenario 1

Embodied carbon intensity (ECI): 42 kgCO₂e/m²
Scenario 1

Lowest Retrofit Cost Payback Period

233 years

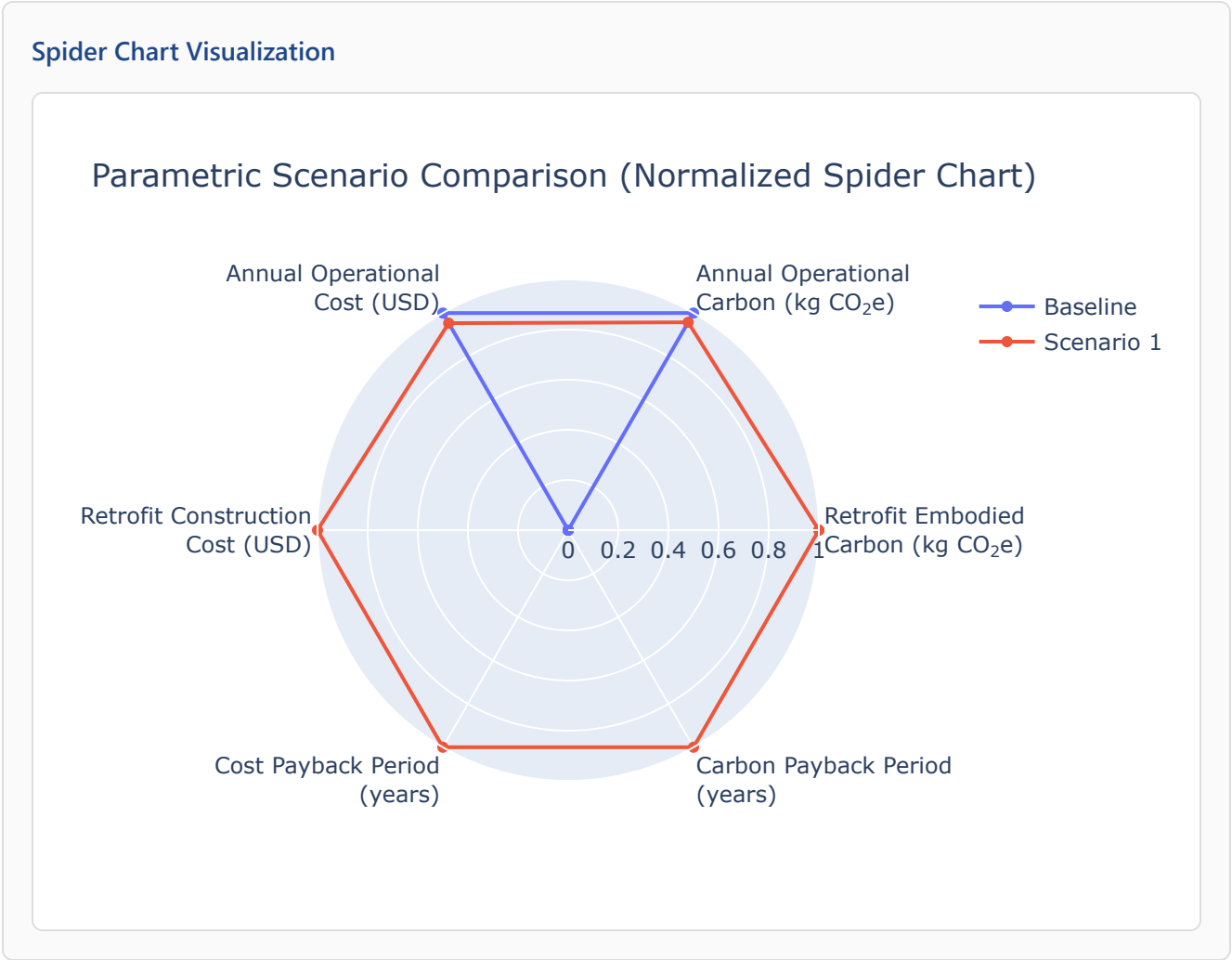
Scenario 1

Lowest Retrofit Carbon Payback Period

24 years

Scenario 1

Comparative Visualizations between Renovation Scenarios



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1		233 yrs	Scenario 1		24 yrs
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Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

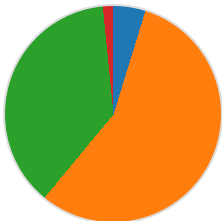
Scenario	Component	Material name	Volume (m ³)	Area (m ²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m ³)	Product Lifetime (years)
Scenario 1	Wall insulation	Mineral Wool Heavy Density Blanket	20	282	N/A	N/A	0.03	103	30
Scenario 1	Roof insulation	Graphite Polystyrene (GPS) Foam Board	116	599	0.19	N/A	0.03	14	75
Scenario 1	Window frame	wood-aluminium window frame	N/A	56	N/A	N/A	N/A	N/A	15
Scenario 1	Window caulking	polyurethane	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 1	Door	honeycomb core steel door	N/A	3.90	N/A	N/A	0.05	482	30
Scenario 1	Door bottom sealing	brush weatherstripping	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 1	Door top/side sealing	jamb weatherstripping	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

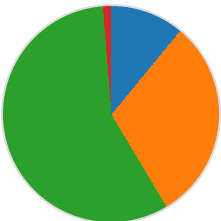
Cost breakdown by retrofit measure



- Wall: \$5,100 (4.8%)
- Roof: \$60,118 (56.2%)
- Window: \$39,986 (37.4%)
- Door: \$1,695 (1.6%)

Total: \$106,899

Carbon breakdown by retrofit measure



- Wall: 2,334 (11.0%)
- Roof: 6,499 (30.5%)
- Window: 12,212 (57.3%)
- Door: 270 (1.3%)

Total: 21,316 kg CO₂e

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	20,894	\$9,841
Scenario 1	21,316	\$106,899	20,001	\$9,383